

Journal Pre-proof

One-step synthesis of novel K⁺ and cyano groups decorated triazine-/heptazine-based g-C₃N₄ tubular homojunctions for boosting photocatalytic H₂ evolution

Jian Yang, Yujun Liang, Kai Li, Gui Yang, Kun Wang, Rui Xu, Xianjun Xie



PII: S0926-3373(19)30999-3
DOI: <https://doi.org/10.1016/j.apcatb.2019.118252>
Reference: APCATB 118252

To appear in: *Applied Catalysis B: Environmental*

Received Date: 9 May 2019
Revised Date: 28 September 2019
Accepted Date: 2 October 2019

Please cite this article as: Yang J, Liang Y, Li K, Yang G, Wang K, Xu R, Xie X, One-step synthesis of novel K⁺ and cyano groups decorated triazine-/heptazine-based g-C₃N₄ tubular homojunctions for boosting photocatalytic H₂ evolution, *Applied Catalysis B: Environmental* (2019), doi: <https://doi.org/10.1016/j.apcatb.2019.118252>

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2019 Published by Elsevier.

One-step synthesis of novel K⁺ and cyano groups decorated triazine-/heptazine-based g-C₃N₄ tubular homojunctions for boosting photocatalytic H₂ evolution

Jian Yang^a, Yujun Liang^{a, *}, Kai Li^b, Gui Yang^a, Kun Wang^a, Rui Xu^b, Xianjun Xie^{b, *}

^a Engineering Research Center of Nano-Geomaterials of Ministry of Education,
Faculty of Materials Science and Chemistry, China University of Geosciences, Wuhan
430074, China

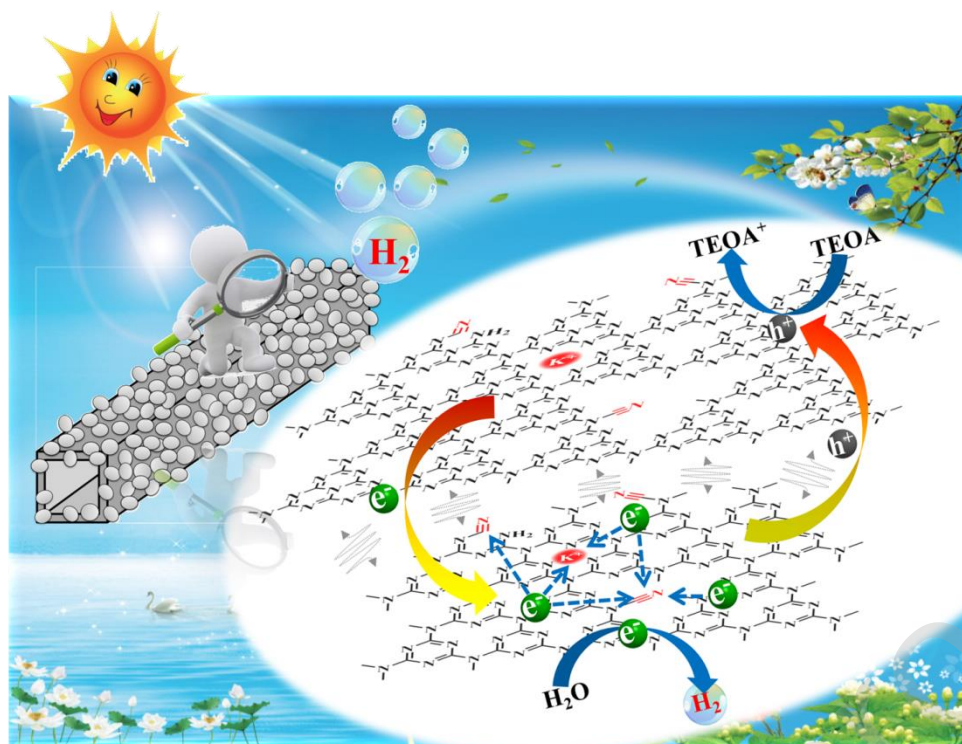
^b State Key Laboratory of Biogeology and Environmental Geology & School of
Environmental Studies, China University of Geosciences, Wuhan 430074, China

***Corresponding author:** Tel.: +86 27 67884814; fax: +86 27 67883733.

E-mail address: yujunliang@sohu.com (Yujun Liang)

xjxie@cug.edu.cn (Xianjun Xie)

Graphical Abstract



Highlights

- A molten salt assisted strategy was designed to construct type- II triazine-/heptazine-based g-C₃N₄ homojunctions.
- The phase and morphology controlling of g-C₃N₄ was achieved by only adjusting the reaction temperature.
- The g-C₃N₄ homojunctions demonstrated 12-times higher hydrogen evolution rate than bulk ones.
- Strong synergistic effects of tubular structure, K⁺ ions, cyano group defects and homojunctions contribute to the enhanced photocatalytic performance.

ABSTRACT

Constructing homojunctions with different morphologies, exposing facets, crystal phases or semiconductor types in a photocatalyst is an essential approach to boost the photoactivity. Herein, the g-C₃N₄ phase homojunctions decorated with cyano groups and K⁺ were designed and successfully synthesized via only adjusting the reaction temperatures within a facile one-step molten salt route. Such unique junctions were constructed through the overgrowth of triazine-based g-C₃N₄ nanoparticles embedded in the surface and inner wall of heptazine-based g-C₃N₄ hollow tubes. Benefited from the multiple advantages of high specific surface area, unique tubular structure, enhanced visible light absorption, and fast charge transfer and separation, the triazine-/heptazine-based g-C₃N₄ homojunctions drastically enhanced the photocatalytic hydrogen production performance, achieving a 2 and 12-fold improvement than the pristine g-C₃N₄ microtube and bulk g-C₃N₄, respectively. This study provides an in-depth insight into the design and fabrication of other g-C₃N₄-based photocatalysts for more efficient solar energy conversion applications.

Keywords: Heptazine-based g-C₃N₄, Triazine-based g-C₃N₄, Homojunctions, Molten salt synthesis, Photocatalytic hydrogen evolution

1. Introduction

Solar driven semiconductor photocatalysis technology is of great research significance and potential application in energy conversion and environmental governance, of which photocatalytic splitting of water is an ideal way to produce clean hydrogen (H₂) fuels in a sustainable manner [1-3]. As a nonmetallic semiconductor material, polymeric graphitic carbon nitride (g-C₃N₄) has aroused considerable interests for H₂ evolution, carbon dioxide reduction, environment purification and selective conversion of organic functional groups under visible light irradiation owing to its suitable band gap, excellent thermal and chemical stability, and cheap raw materials (abundant nitrogen-rich precursors) [4]. Discontentedly, the photocatalytic performance of bulk g-C₃N₄ resultant from the traditional thermal polymerization route is significantly limited by its intrinsic drawbacks of low specific surface area, narrow visible-light response range, and rapid recombination rate of photogenerated carriers [5-7]. Therefore, unremitting efforts have been committed to solve the above deficiencies for boosting the visible light photocatalytic activity of g-C₃N₄. Realization of the adjustments of morphology and structure depends on the development of synthetic methodologies to increase specific surface area, adjust the band structure and ameliorate carrier transport mechanism. Hitherto, various

nano-scale morphologies including nanorods [8], nanosheets [9,10], nanotubes [11-18], nanoporous [19], seaweeds [20], and hollow spheres of g-C₃N₄ [21] have been fabricated and all of them exhibited excellent visible light photocatalytic activity than the bulk one. Notably, the g-C₃N₄ tubular nanostructure could not only shorten the distance of photoinduced carriers moving from the bulk phase to the surface, but also provide large surface area and expose abundant active sites for photocatalytic reaction. Moreover, the hollow structure could also enhance the light absorption by multiple light scattering/reflection effect from the inner cavity, thus inducing the favorable photocatalysis.

Generally, the reported synthesis methods of tubular g-C₃N₄ mainly included soft template, hard template, self-assembly and hydrothermal routes [11-16]. Unfortunately, there were some obstacles in the mentioned means as low efficiency, harsh synthetic procedures, complicated manipulation, and specific requirement of equipment and/or large consumption of hazardous organic agents. Thus, exploring an alternative method for the synthesis of tubular g-C₃N₄ is still of technological as well as scientific significance. As a sustainable, cost-effective and feasible method, molten salt (MS) synthesis has been considered as a prospective approach for the morphology controlled and surface modification of functional nanomaterials, in which the eutectic salt compositions could be as a highly reactive medium to accelerate the diffusion speeds of reacting constituents and reduce the synthesis temperature [22-24]. Recently, MS method has been utilized to prepare K-Na co-doped g-C₃N₄ [25], triazine-based g-C₃N₄ [26], tetragonal g-C₃N₄ hollow tubes [27], water-dispersible g-C₃N₄

nanoseaweeds [28], g-C₃N₄-MoS₂ hybrids [29], and so forth. Simultaneously, introducing functional groups (i.e., -OH [30], -CN [31-33]), nonmetallic elements (i.e., Br [34], S [35], P [17]), transition metal ions (i.e., Fe [36], Mn [37]) and alkali metal ions (i.e., Na [38], K [6]) into the framework of g-C₃N₄ could further improve the light absorption, adjust the band structure, and act as the strong electron-withdrawing groups with the purpose of driving the transfer kinetics of photoinduced charge carriers. Therefore, the establishment of tubular g-C₃N₄ decorated with the above functional groups is likely to be achieved in a highly reactive medium.

By considering a much related topic, the fabrication of heterojunctions is also demonstrated as one of the most forward-looking approaches to further boost the visible light photocatalytic performance of g-C₃N₄. Recently, Zhang et al. constructed the 2D/2D BiOBr/CDs/g-C₃N₄ Z-scheme heterojunction photocatalyst with carbon dots as solid-state electron mediators for enhanced visible and NIR photocatalytic activity [39]. Cui's group reported that 1T-MoS₂ quantum dots (QDs) could act as co-catalysts in assisting the photocatalytic H₂ evolution to form heterostructures with g-C₃N₄ nanosheets [40]. Yang et al. fabricated the fish scale-like g-C₃N₄/Ag₃PO₄ heterojunction photocatalyst with the excellent oxygen-evolving activity under light-emitting diode (LED) illumination [41]. In particular, compared with heterojunctions built from different components, the homojunctions fabricated by one photocatalyst with different morphologies, exposed facets, crystal phases or semiconductor types, etc., could deliver the continuity of band bonding and keep the movement of charge carriers across the interface more efficiently [42-47]. For

instance, Ye's group utilized n- and p-type defect-modified g-C₃N₄ to fabricate p-n homojunctions for enhancing photocatalytic H₂ production [33]. Mesoporous g-C₃N₄/g-C₃N₄ nanosheets laminated homojunctions were synthesized through template-calcination strategy using melamine and amino cyanamide as co-precursors [45]. Two dimensional/two dimensional g-C₃N₄ homojunctions photocatalyst was also obtained using various g-C₃N₄ nanosheets as building blocks [46]. Thus far, the reported g-C₃N₄ homojunctions were mainly manifested on different morphologies or semiconductor types by multi-step methods. Fundamentally, g-C₃N₄ has two major allotropes with the dissimilar phases: one is the most studied heptazine-based g-C₃N₄ and the other is the triazine-based g-C₃N₄, which are composed of two basic tectonic units of tri-s-triazine (or heptazine) ring and s-triazine ring, respectively (Fig. 1a and b) [1]. In fact, design of isotype triazine-/heptazine based g-C₃N₄ homojunctions was anticipated to be an effective tactic for considerably augmenting the photocatalytic activity [7,47]. In consideration of the above analysis, presumably, the advantages of electron-withdrawing groups and homojunctions are incorporated into g-C₃N₄ tubular structures through the convenient MS might increase energy conversion efficiency to a greater extent.

In this contribution, we reported a facile one-step MS route for the preparation of tubular triazine-/heptazine-based g-C₃N₄ intergrowth homojunctions which were composed of triazine-based g-C₃N₄ nanoparticles and heptazine-based g-C₃N₄ hollow microtubes. The formation of closely contacted junctions was achieved by only adjusting the reaction temperatures. Meanwhile, we interestingly found that the cyano

groups and K^+ ions could also be introduced in the framework of g-C₃N₄. The strong synergistic effects of these factors could commendably enlarge the specific surface area, improve the light absorption and accelerate the charge separation, resulting in a drastically enhanced visible light photocatalytic H₂ evolution performance. Furthermore, the comprehensive photocatalytic mechanism was also proposed and discussed in detail.

2. Experimental section

2.1. Preparation of photocatalysts

All reagents used in this work were of analytical reagent grade and without further purification. In a typical process, 1 g melamine was fully ground together with 5 g of eutectic mixture of LiCl and KCl (in molar ratio of 59:41, with the eutectic point of 352 °C [48]) in an agate mortar more than 15 min to obtain a homo-disperse mixture. After that, the grinded mixture was placed in an alumina crucible which covered and wrapped with aluminium-foil paper, and then annealed in a muffle furnace at different temperatures (450 °C, 500 °C, 550 °C and 600 °C) for 4 h with a heating rate of 5 °C/min. After cooling down naturally to room temperature in the furnace, the obtained samples were washed with hot deionized water and ethanol for several times, and then dried at 80 °C overnight. Finally, these as-prepared samples were uniformly denoted as MS-CN. In detail, the samples synthesized at different temperatures were labeled as MS-450, MS-500, MS-550, and MS-600, respectively.

For the sake of comparison, the bulk g-C₃N₄ was also prepared by a conventional thermal polymerization route. Briefly, 1 g melamine was placed in the crucible with a ceramic lid and heated at 520 °C for 4 h with a heating rate of 5 °C/min. The resultant product was simply designated as B-CN.

2.2. Characterization

Powder X-ray diffraction (XRD) data of the as-prepared samples were collected on an X-ray diffractometer (D8-FOCUS, Bruker, Germany) with Cu K α radiation ($\lambda = 1.5418 \text{ \AA}$). Fourier transform infrared (FT-IR) spectra in the wavenumber range of

4000-500 cm^{-1} were recorded on a Nicolet 6700 spectrometer with KBr pellet (Thermo scientific, USA). The morphology and microstructure observation were examined by field emission scanning electron microscope (FE-SEM, SU8010, Hitachi, Japan) and transmission electron microscope (TEM, Tecnai G2 T20, FEI, America). Meanwhile, the energy-dispersive X-ray spectroscopy (EDS) spectra and elemental mapping images were collected in an EDAX Genesis, which was attached to the FE-SEM. The N_2 adsorption/desorption isotherms at 77 K were conducted on an automatic surface area analyzer (MicroActive for ASAP 2460, America), and the specific surface areas of as-prepared samples were measured using the Brunauer-Emmett-Teller (BET) method. X-ray photoelectron spectroscopy (XPS) measurement was performed on a Thermo Scientific Escalab 250X spectrometer equipped with an Al $\text{K}\alpha$ X-ray source. UV-vis diffuse reflectance spectra (DRS) were tested on an UV-2550PC spectrophotometer (Shimadzu Corporation, Japan), using BaSO_4 as the reflectance standard. The photoluminescence (PL) spectra of photocatalysts were measured by a fluorescence spectrometer (Fluoromax-4P, Horiba Jobin Yvon, New Jersey, USA), which was equipped with a 150 W xenon lamp as the excitation source. The photoluminescence decay curves were performed from the Fluorescence Spectrometer (FLS 920, Edinburgh Instruments, Livingston, UK).

2.3. Photocatalytic H_2 evolution

The photocatalytic H_2 production experiments were conducted in a sealable quartz reactor with top visible light irradiation. A 300 W Xe lamp with a cut-off filter ($\lambda > 420 \text{ nm}$) was used as the visible light source, and the fixed light-to-liquid distance was

10 cm. In detail, 100 mg of the as-prepared sample was dispersed in 100 mL aqueous solution containing 10 vol% triethanolamine (TEOA) as the sacrificial agent for the holes, and 3 wt% Pt as the cocatalyst was loaded on the surface of catalyst by the conventional in situ photodeposited method using $\text{H}_2\text{PtCl}_6 \cdot 6\text{H}_2\text{O}$. The mixture solution was continuously stirred by a magnetic stirrer. Before visible light irradiation, the reactant system was purged by N_2 for 30 min to remove air. The amount of H_2 evolution was analyzed by a gas chromatograph (GC7900, Techcomp) equipped with a thermal conductivity detector.

The apparent quantum efficiency (AQE) of the as-prepared photocatalyst was determined by photocatalytic water-splitting H_2 evolution under the monochromatic light irradiation (a series of bandpass filters were used, $\lambda = 420$ nm, 450 nm and 500 nm) and calculated from the equation [22]:

$$\begin{aligned} \text{AQE} &= \frac{2 \times \text{number of hydrogen molecules}}{\text{number of incident photon}} \times 100\% = \frac{2 \times M \times N_A}{\frac{E_{\text{total}}}{E_{\text{photon}}}} \times 100\% \\ &= \frac{2 \times M \times N_A}{\frac{S \times P \times t}{h \times \frac{c}{\lambda}}} \times 100\% = \frac{2 \times M \times N_A \times h \times c}{S \times P \times t \times \lambda} \times 100\% \end{aligned}$$

where M represents the amount of H_2 molecules (mol), N_A is Avogadro constant ($6.022 \times 10^{23} \text{ mol}^{-1}$), h is the Planck constant ($6.626 \times 10^{-34} \text{ J s}$), c is the speed of light ($3 \times 10^8 \text{ m s}^{-1}$), S stands for the irradiation area (cm^2), P is the power density of the incident monochromatic light (W cm^{-2}), λ is the wavelength of the irradiated monochromatic light (m) and t (s) indicates the photoreaction time.

2.4. Electrochemical measurements

The photocurrent response spectroscopy, electrochemical impedance spectroscopy (EIS) and Mott-Schottky plots were collected on a standard three-electrode electrochemical analyzer (CHI760E, Shanghai). The as-prepared sample coated onto the FTO electrode was served as the working electrode, an Ag/AgCl (saturated KCl) as the reference electrode, and a Pt foil as the counter electrode. The Na₂SO₄ (0.1 M) solution and 300 W Xe lamp were employed as electrolyte and light source, respectively. The working electrodes were prepared according to previously reported method [33]: 5 mg of photocatalyst powder and 10 μ L of Nafion solution (5 wt%) were dispersed into 100 μ L isopropanol and 300 μ L water, and then ultrasonic treatment for 1 h to prepare a homogeneous catalyst colloid. Finally, the resultant catalyst slurry was coated onto the precleaned FTO glass surface with an active area of ca. 1.00 cm $\times$ 1.00 cm and then dried in air.

3. Results and discussion

Typical XRD patterns of the as-prepared samples are depicted in Fig. 1c. In the case of MS-450 sample, two distinct diffraction peaks appeared at 2θ value of 13.0° and 27.8°, which could be indexed to the (100) and (002) diffraction planes of the typical g-C₃N₄ phase based on tri-s-triazine building blocks (i.e., heptazine-based g-C₃N₄) [49]. MS-450 and B-CN appeared the similar characteristic diffraction peaks (Fig. S1), suggesting that the MS method could effectively reduce the synthesis temperature of g-C₃N₄. Besides, the XRD pattern of MS-600 displayed a series of strong diffraction

peaks with 2θ at 12.1° , 21.0° , 24.4° , 26.7° , 29.3° and 32.2° , which could be ascribed to the poly (triazine imide) (PTI) phase (i.e., triazine-based $g\text{-C}_3\text{N}_4$), corresponding to the major (100), (110), (200), (002), (102) and (210) planes [27,47]. Comparatively, it was interestingly observed that the MS-500 and MS-550 samples showed a compound of heptazine-based $g\text{-C}_3\text{N}_4$ and triazine-based $g\text{-C}_3\text{N}_4$ phases, and no other peaks from possible impurities were detected in XRD patterns, implying that the triazine-/heptazine-based $g\text{-C}_3\text{N}_4$ homojunctions were successfully synthesized via a one-step MS pathway. In addition, compared with MS-450, the diffraction peak from the (002) plane of the heptazine-based $g\text{-C}_3\text{N}_4$ phase revealed a slight right-shift in MS-500 and MS-550 samples, which suggested a decreased interlayer distance between two adjacent unit layers [11]. Moreover, we also found that no diffraction peaks of $g\text{-C}_3\text{N}_4$ could be observed when the heating temperature was 400°C (Fig. S2), and no products was yielded when the heating temperature up to 650°C (Fig. S3). Consequently, it could be concluded that the synthesis temperature of $g\text{-C}_3\text{N}_4$ (single phase or mixed phases) by the MS method was ranged from 450°C to 600°C , and the proportion of triazine-based $g\text{-C}_3\text{N}_4$ phase was rose with the increasing reaction temperature.

The corresponding FT-IR spectra were recorded to examine the functional groups and molecular structure of as-prepared samples, as shown in Fig. 1d. It was clear to observe that the MS-450 and B-CN revealed the similar FT-IR vibration modes, suggesting that the heptazine-based $g\text{-C}_3\text{N}_4$ synthesized with molten salt method maintained a chemical core structure which was similar to B-CN. The obvious

adsorption peak at 806 cm^{-1} was corresponded to the ring-sextant out-of-plane bending vibration of s-triazine units [26], and the broad peaks in the range of $3000\text{-}3600\text{ cm}^{-1}$ were assignable to the absorbed H_2O molecules and terminal amino groups [50]. Additionally, the most intense absorption bands in the $1200\text{-}1700\text{ cm}^{-1}$ region were originated from the stretching vibration modes of C-N heterocycles [38]. In particular, the absorption bands of MS-CN located at around 1278 cm^{-1} , were significantly higher than that of B-CN (1243 cm^{-1}), which probably because the few K^+ ions entrapped into the framework of g- C_3N_4 and affected the characteristic vibrations of g- C_3N_4 matrix. Besides, a distinct peak located at 2175 cm^{-1} was found from the spectra of the MS-CN, which was ascribed to the terminal cyano groups ($-\text{C}\equiv\text{N}$), caused by the catalytic pyrolysis of g- C_3N_4 that lost ammonia on the surface in the presence of alkali metal chloride [28,51]. Compared with B-CN, the other new peaks of MS-CN samples located at about 1172 cm^{-1} and 997 cm^{-1} were attributed to the hydroxyl group, which might be caused by the enlarged specific surface area [30]. It was worth mentioning that the peak at 887 cm^{-1} was arose from the characteristic bending vibration of the extended conjugated CN-networks with tri-s-triazine units [27,52], so this signal appeared in the FT-IR spectra of MS-450 and B-CN, but absented from the spectrum of MS-600. Meanwhile, a characteristic frequency at about 1382 cm^{-1} was present in MS-600 due to the condensed C-N heterocycles of triazine-based g- C_3N_4 , which could be used to distinguish it from heptazine-based g- C_3N_4 [47]. More importantly, both MS-500 and MS-550 samples exhibited the

characteristic peaks of 887 and 1382 cm^{-1} , further confirming the successful evolution of the triazine-/heptazine-based $\text{g-C}_3\text{N}_4$ homojunctions.

The microstructure and morphology of MS-CN were characterized with FE-SEM. Unlike the B-CN which was synthesized by the traditional thermal polymerization route (Fig. S4), MS-450 (heptazine-based $\text{g-C}_3\text{N}_4$) displayed a typical tubular morphology with the relative smooth surface (Fig. 2a), and this distinctive structure usually keeps a larger specific surface area. It was notable to observe that the sample synthesized at 500 $^{\circ}\text{C}$ exhibited a mixed-phase structure constituted by hollow tube and the new formed nanoparticles on the surface of tube (Fig. 2b). According to the XRD result, the MS-500 contained both heptazine-based $\text{g-C}_3\text{N}_4$ and triazine -based $\text{g-C}_3\text{N}_4$ phases, so the attached nanoparticles would be the triazine-based $\text{g-C}_3\text{N}_4$. As could be seen in Fig. 2c, the surface and inner walls of tube became rougher and the number of nanoparticles increased when the thermal treatment temperature continually increased to 550 $^{\circ}\text{C}$, which was consistent with the XRD results. Comparatively, MS-600 sample was composed of textured triazine-based $\text{g-C}_3\text{N}_4$ nanoparticles (Fig. 2d). Although some hollow tubes tended to deteriorate after further increasing calcination temperature to 600 $^{\circ}\text{C}$, the residual tubing structures still could be observed. In order to more clearly reflect the microstructure and morphology of MS-CN, we provided others SEM images which included more $\text{g-C}_3\text{N}_4$ tubular homojunctions structures in Fig. S5. Moreover, the EDS spectra results confirmed the existence of K in the MS-CN, demonstrating that K^+ ions were successfully doped (Fig. S6). As shown in Fig. 2(e-h), the elemental mapping images revealed that C, N

and little K elements were homogeneous distributed in the exceptional tubular structure.

In order to gain a deeper insight into the detailed structures of triazine-/heptazine-based g-C₃N₄ homojunction, the TEM images of MS-550 are provided in Fig. 3a and b. It could be seen clearly that some nanoparticles with the size of ca. 20-30 nm were successfully embedded in the surface and inner walls of hollow tube. Although the as-prepared sample displayed a weak crystallinity, two different kinds of lattice fringes could be observed at the interface of heptazine-based g-C₃N₄ and triazine-based g-C₃N₄ (Fig. 3c). One lattice spacing with $d = 0.320$ nm was corresponded to the inter-layer structural packing, indexing for the (002) plane of heptazine-based g-C₃N₄, and the other with $d = 0.336$ nm was matched the (002) crystallographic plane of triazine-based g-C₃N₄ [47,62]. These typical lattice spacing were well consistent with the XRD results. In addition, this kind of embedded-homojunction synthesized by one-step MS method was more closely bound up with each component than the contacted-heterojunction. Hence, it was expected that the formed homojunction with the intimately contacted interface would much favorable for the separation of the photogenerated charge carriers. The typical N₂ adsorption-desorption isotherm curves of B-CN and MS-CN are presented in Fig. 3d, the isotherms for all samples belong to the type IV, demonstrating the existence of mesoporous (2-50 nm). It could be perceived from the inset of Fig. 3d that the BET surface areas of all MS-CN were much larger than that of B-CN. Moreover, the BET surface area of the MS-550 could attain approximately $106 \text{ m}^2 \text{ g}^{-1}$, which was about

18 times higher than that of the B-CN ($6 \text{ m}^2 \text{ g}^{-1}$). In contrast, the BET surface area of MS-600 decreased, which might be attributable to the damaged hollow tubular structure and the agglomerated nanoparticles. Accordingly, it could be concluded that the optimization of calcination temperature was imperative to achieve the formation of homojunctions with outsized specific surface area, which could deliver lavish surface active sites for adsorption and surface reaction.

A plausible schematic illustration of the phase transformation and morphology evolution from heptazine-based $\text{g-C}_3\text{N}_4$ microtubes to triazine-based $\text{g-C}_3\text{N}_4$ nanoparticles is illustrated in Fig. 4. In the initial stage, the whole system was embraced in a highly reactive circumstance at molten state, because the LiCl-KCl eutectic mixture possessed a eutectic point at about 352°C based on the binary phase diagram (Fig. S7). When the temperature reached to 389°C , some melamine molecules could form the polymeric melem [47]. Liu et al. also reported that the melamine and melem units could coexist in the presence of the combination of LiCl and KCl, and the eutectic mixture would provide a special high-temperature liquid media to improve the polymerization reaction of melamine and melem for the formation of triazine-based fragments and heptazine-based fragments, respectively [47]. In our work, no other peaks from triazine-based $\text{g-C}_3\text{N}_4$ were detected in the XRD pattern of MS-450, implying that the triazine-based fragments derived from the melamine were absence in the reaction system. The unreacted melamine would be completely transformed into melem species with the reaction temperature rising to 450°C . Namely, the polymerization reactions of heptazine-based fragments continued

and finally formed the heptazine-based g-C₃N₄. The stepwise ionization of molten salt could offer adequate time for the structure assembling and directly impact the surface property and minimize the surface energy to stabilize the crystal lattice [53]. As Tian et al. reported, the formation of carbon nitride hollow tubes could be expressed as a salt-assisted self-assembly mechanism in the molten salt medium of LiCl-KCl eutectic mixture [27]. Additionally, Xing's group found that some tri-s-triazine (melem) units consisted of C-N bonds could be "etched" by further thermal oxidation [10]. Therefore, the increased calcination temperature would initiate the gradual decomposition of heptazine-based g-C₃N₄ and the surface roughening of tubular structure. Some triazine-based fragments might be obtained and adsorbed on the surface of microtubes from the in-situ decomposition of heptazine-based g-C₃N₄ at the temperature above 450 °C. Meanwhile, the polymerization reaction of triazine-based fragments for the formation of triazine-based g-C₃N₄ would be accelerated in the high-temperature liquid medium, leading to the overgrowth of triazine-based g-C₃N₄ nanoparticles embedded in the surface and inner wall of heptazine-based g-C₃N₄ hollow tubes. Ultimately, the heptazine-based g-C₃N₄ microtubes were fully decomposed at 600 °C, resulting in the weight loss and volume shrinking of products (Fig. S3), and consequently, a destroyed tubular structure which assembled by triazine-based g-C₃N₄ nanoparticles was yielded. Consequently, in this study, the triazine-based g-C₃N₄ were originated from the decomposition of heptazine structure rather than the thermally pyrolyzed and polycondensed process of melamine. This

temperature-induced in-suit transformation and growth could afford an achievable pathway for the controllable synthesis of phase junctions of g-C₃N₄.

XPS was carried out to further investigate the surface elemental composition and specific bonding in B-CN and MS-550 samples. As plotted in Fig. 5a, the survey spectra suggested that the C, N and O elements could be detected in both B-CN and MS-550, in which the C and N elements were originated from the g-C₃N₄, and the elemental O was belonged to the adsorbed oxygen species on the sample surface during the polymerization process [54]. Therefore, the appearance of strong intensity of O 1s peak in MS-550 was due to the larger specific surface area. The high resolution XPS spectra of C 1s region could be divided into three peaks located at approximately 284.5, 286.9 and 288.0 eV, respectively (Fig. 5b). The binding energy peak at 284.5 eV was assigned to the surface adventitious carbon, while the peak at 288.0 eV was corresponding to the typical sp²-bonded carbon in (N=C-(N)₂) coordination [55]. The other peak at 286.9 eV was related to the C-OH group [56]. In comparison with B-CN, the peak at 286.9 eV of MS-550 became stronger, indicating the hollow tubular structure had the superior specific surface area, which was consistent with the BET results. Meanwhile, the high resolution XPS spectra of N 1s region could be fitted to four peaks (Fig. 5c). The primary peak at 398.4 eV revealed the nitrogen atoms in the sp² C-N=C involved both in the s-triazine rings and heptazine rings, the peak at 399.7 eV was originated from the bridging nitrogen atoms N-(C)₃, while the peaks at 400.4 and 404.3 eV were attributed to the C-N-H functional groups and π excitation, respectively [33,57]. It was worth noting that the peak area of

$\text{N}-(\text{C})_3$ of MS-550 sample was much smaller than that of B-CN. The phenomenon could be explained by the fact that the signal of $\text{N}-(\text{C})_3$ was mainly came from the heptazine rings of heptazine-based $\text{g-C}_3\text{N}_4$, and the $\text{N}-(\text{C})_3$ groups were not contained within the s-triazine rings of triazine-based $\text{g-C}_3\text{N}_4$. Hence, it could be inferred again that the coexistence of heptazine-based $\text{g-C}_3\text{N}_4$ and triazine-based $\text{g-C}_3\text{N}_4$ phases in MS-550 sample. Additionally, the K 2p spectrum of MS-550 in Fig. 5d demonstrated the two major peaks at 295.3 and 292.6 eV, which could be assigned to the typical binding energy of K^+ ions. Interestingly, there were no obvious signals of Cl-related species appeared, hinting that K^+ ions might be entrapped into the framework of the $\text{g-C}_3\text{N}_4$ structure during the crystal growth process rather than existed in the form of KCl. Likewise, the binding energies of C 1s and N 1s of MS-550 were lower than that of B-CN, which was owing to the strong interplay between K ions and N atom [58]. More importantly, the atomic radii of N (65 pm) and C (70 pm) were much smaller than that of K^+ (about 150 pm), and the maximum in-planar distance of the nitride pores was 0.71 nm [59], which was enough to accommodate the K ions. Therefore, the interstitial doping was more easily occurred than substitutional doping in molten salt medium. Namely, K^+ ions were more likely to embed into the C-N rings with threefold N-bridge linking the s-triazine units or tri-s-triazine units. Besides, it was known that $\text{g-C}_3\text{N}_4$ was a covalent compound, so K^+ ions could not be doped as an ion state in a substitutional site. The contents of K^+ ions in MS-CN were determined by EDS (Table S1), and it could be observed that the K^+ ions gradually decreased with the increase of the proportion of triazine-based $\text{g-C}_3\text{N}_4$ phases in the hybrids. This was

because of the fact that the triangular cage enclosed by three C_6N_7 rings in heptazine-based $g-C_3N_4$ was larger than the one enclosed by three C_3N_3 rings in triazine-based $g-C_3N_4$, and the interstitial doping of some K^+ ions in triazine-based $g-C_3N_4$ might be hindered.

Light absorption efficiency is one of the imperative factors that affect the photocatalytic performance of the catalysts. To assess the photoabsorption behaviors of as-prepared photocatalysts, the UV-vis diffuse reflectance spectra (DRS) were conducted, which are illustrated in Fig. 6a. It was noteworthy that the MS-450 sample possessed the similar absorption edge with B-CN, but both the ultraviolet and visible light absorptive capacities were enhanced corresponding to the introduction of cyano groups and K^+ ions. At the same time, the MS-600 displayed strong absorption in the ultraviolet region, and the absorbance in the visible region was negligible. Interestingly, the absorption edges of MS-500 and MS-550 samples exhibited a distinct red-shift, which could be ascribed to the interactions between the tightly-embedded surface of triazine-based $g-C_3N_4$ and that of heptazine-based $g-C_3N_4$ (via the formed homojunctions). The extended light absorption of composites was also consistent with the colors change (Fig. S8). Furthermore, the band gap energies (E_g) of all samples could be estimated by the Kubelka-Munk method, as displayed in Fig. 6b. According to the intercept of the tangents to the plots of $(\alpha h\nu)^{1/2}$ vs. photon energy ($h\nu$), the E_g of B-CN, MS-450 and MS-600 were determined to be 2.61, 2.58 and 2.81 eV, respectively. Clearly, MS-500 and MS-550 samples exhibited a narrower

E_g than those of single g-C₃N₄ phases, so more photoinduced electrons and holes would be yielded in homojunctions under visible light excitation.

The positions of valence band (VB) top and conduction band (CB) bottom are the key characteristics of band structure responsible for the photocatalytic properties. Mott-Schottky plots of B-CN, MS-450 and MS-600 were collected under three different frequencies (500, 1000 and 1500 Hz), as shown in Fig. S9. According to the tangent intersection points of Mott-Schottky plots (Fig. 6c), the flat-band potentials of B-CN, MS-450, and MS-600 were estimated to be -0.67, -0.89, and -0.71 V (vs. Ag/AgCl), corresponding to -0.45, -0.67 and -0.49 V (vs. NHE), respectively. At the same time, the positive slopes of the fitted curves indicated that all of the samples were n-type semiconductors [60]. Generally, the CB position of n-type semiconductor was close to the flat-band potential [61]. Combined with the values of E_g , the VB positions were also estimated to be 2.16, 1.91 and 2.32 eV for B-CN, MS-450, and MS-600, respectively. Therefore, a schematic vision of the CB and VB potentials for the samples is depicted in Fig. 6d. Evidently, the negative-shift of CB position in MS-450 would result in the stronger photocatalytic reductive ability of photogenerated electrons in terms of thermodynamic requirements for photocatalytic reactions. Consequently, it was sagacious to support the conclusion that the successful combination of two components of g-C₃N₄ phases could greatly boost visible light absorption capacity and narrow the band gap of g-C₃N₄, which would be conducive to further improve the photocatalytic performance.

It is well known that the transfer and separation efficiencies of photogenerated electron-hole pairs play the pivotal role in determining the photocatalytic activity. The PL spectra of MS-CN were recorded under the excitation wavelength of 350 nm (Fig. 7a). It could be seen that PL spectrum of the MS-600 presented a blue shift of about 23 nm compared with MS-450 owing to the larger band gap of triazine-based g-C₃N₄, which was consistent with the UV-vis DRS result. Compared with the MS-450 and MS-600, the dramatically quenched emission intensities of MS-500 and MS-550 could be observed, and the MS-550 exhibited a minimum PL intensity, suggesting that the remarkable enhancement appeared in the separation of the photoexcited carriers with the formation of g-C₃N₄ homojunctions. Besides, the red-shifts in the PL maximum were observed in MS-500 and MS-550, which was associated with the decreased E_g . Moreover, the PL intensity of MS-450 was obviously much lower than that of B-CN (Fig. S10), such a behavior was believed to be driven by the cyano group defects and interstitial doping of K⁺ ions. Meanwhile, the thin walls of the tubular structure could also reduce the transmission distance of carriers.

Normally, the photocurrent could also provide an efficient evidence to reveal the transfer property of the photogenerated charges. Fig. 7b displays the transient photocurrent responses of MS-450, MS-500, MS-550 and MS-600 samples. As expected, the photocurrent intensities of MS-500 and MS-550 were all higher than that of MS-450 and MS-600. Especially, the MS-550 exhibited the highest photocurrent, hinting it hold the fastest charge transfer rate and the longest lifetime of photo-induced carriers. In accordance with the photocurrents, EIS Nyquist plots also

described the similar results (Fig. 7c). The diameter of the arc radius on the EIS Nyquist plot of MS-550 was much smaller than that of MS-450 and MS-600, which indicated a smaller interface resistance for MS-550. In order to gain a deeper insight into the photo-physical characteristics of photo-induced charge carriers, the ns-level time-resolved fluorescence decay spectra were also carried out and are shown in Fig. 7d. Relative to MS-450 (3.782 ns) and MS-600 (3.125 ns), MS-500 and MS-550 exhibited the reduced average fluorescence lifetimes (2.253 ns and 1.680 ns, respectively), which was illustrated as the enhanced electron-hole separation and transfer encouraged by the closely contact junctions effect. When considering the photocatalytic H₂ evolution performance, many reports indicated that the shorter the lifetime, the larger the H₂ photo-generation rate, because the possibility of charges to react with the electron or hole captures was increased [7, 47]. In summary, these results incontrovertibly indicated that the elongated lifetime, enhanced generation and separation of photogenerated electrons-holes could be achieved by fabricating the triazine-/heptazine-based g-C₃N₄ homojunctions decorated with the cyano group defects and K⁺ ions interstitial doping, which would induce favorable photocatalysis.

On the basis of confirming the structure, light absorption and charge separation efficiency of the as-prepared catalysts, the photocatalytic activities of these samples were evaluated by decomposing water to produce H₂ under visible light irradiation. As illustrated in Fig. 8a, the B-CN prepared by traditional thermal polymerization method exhibited the poor H₂ production rate of about 54 $\mu\text{mol g}^{-1} \text{h}^{-1}$. Although B-CN and MS-450 all belonged to the heptazine-based g-C₃N₄, MS-450 largely

enhanced H₂ evolution activity than that of B-CN. The superior performance could be ascribed to the high specific surface area of tubular structure and the introduction of K⁺ ions and cyano groups defect in the g-C₃N₄ framework. Evidently, the MS-660 sample displayed barely any photocatalytic performance due to the unsatisfactory visible light absorbance. Moreover, it was worth noting that the H₂ evolution rate of MS-500 (480 $\mu\text{mol g}^{-1} \text{h}^{-1}$) was higher than those of MS-450 and MS-600. Incredibly, MS-550 performed the best photocatalytic performance with the H₂ evolution rate reaching 661 $\mu\text{mol g}^{-1} \text{h}^{-1}$, which was approximately 12 times higher than that of B-CN. The results directly certified the positive effects of the construction of triazine-/heptazine-based g-C₃N₄ homojunctions decorated with the K⁺ ions interstitial doping and cyano group defects on promoting the photocatalytic activity. Fig 8c is the wavelength-dependent AQE for H₂ evolution over MS-550 under different monochromatic light irradiation, the calculated AQE was about 2.88% at 420 nm, 0.59% at 450 nm, and 0.16% at 500 nm, which was corresponded well to its optical absorption. This further supported that the H₂ evolution was indeed a light-induced photocatalytic reactions. Regarding the stability, MS-550 was employed as the representative sample to investigate the stability and reusability of photocatalytic H₂ evolution. As we could see in Fig. 8d, there was no obvious decay in the H₂ production rate after four consecutive cycles, which might owing to the good compatibility of K⁺ ions and cyano groups in the g-C₃N₄ frameworks. Furthermore, the K⁺ ions content, XRD, and FR-IR spectra after cycling tests revealed almost no change within 16 h of photocatalytic reaction (Fig. S11 and Table S1), clearly

demonstrating the superior long-term stability of triazine-/heptazine-based g-C₃N₄ homojunctions.

According to the above investigations, the tentative mechanism for charge separation process of the triazine-/heptazine-based g-C₃N₄ homojunction is proposed in Fig. 9. As is known to all, the formation of heterojunction is an essential bridge for the separation and transfer of photogenerated carriers. Under visible light irradiation, the electrons at the VB of heptazine-based g-C₃N₄ were excited to CB and concurrently, leaving behind the generated holes in the VB. In the case of triazine-based g-C₃N₄, there was only a few electrons could be excited owing to the weak visible light absorbability. After the closely junctions formed between triazine-based g-C₃N₄ and heptazine-based g-C₃N₄, the photogenerated electrons on the CB of heptazine-based g-C₃N₄ would be immediately injected to the CB of triazine-based g-C₃N₄, since the CB and VB of triazine-based g-C₃N₄ were relatively lower than that of heptazine-based g-C₃N₄. Simultaneously, the photoinduced holes on the VB of triazine-based g-C₃N₄ would migrate to the VB of heptazine-based g-C₃N₄. As a consequence, the efficient separation of electrons and holes could be achieved. Besides, the tubular structure could not only assure the brilliant bulk-to-surface and interfacial charge transfer abilities, but also provide the large surface area and expose abundant active sites to adsorb H⁺ and promote mass transfer. Meanwhile, cyano group defects could also play the significant role of the photocatalytically relevant sites [31]. In addition, the hollow structure could also enhance the light absorption by multiple light scattering/reflection effects from the

inner cavity (Fig. 9a). Moreover, the K^+ ions doped into the framework of g-C₃N₄ would cause the non-uniform spatial charge distribution, which was beneficial for the separation of the photoinduced electron-hole pair. On the other hand, Chen et al. [63] found that the electron density of cyano group was obviously higher than any other parts on the g-C₃N₄ surface by the visualization of electron localization function (ELF), indicating that the cyano groups in the framework of g-C₃N₄ could serve as the efficacious electron-withdrawing groups [64]. Therefore, a moderate amount of produced cyano groups defects and K^+ ions in the framework of g-C₃N₄ not only improve the light absorption but also boost the photogenerated charges separation and suppress their recombination (Fig. 9b). Understandably, it was reasonable to conclude that the strong synergistic effects of tubular structure, cyano group defects, K^+ ions doping and triazine-/heptazine-based g-C₃N₄ homojunctions could enhance light absorption and improved photoexcited charge carrier separation, giving rise to the excellent photocatalytic H₂ evolution.

4. Conclusions

To summarize, we have successfully prepared triazine-/heptazine-based g-C₃N₄ homojunctions with the cyano group defects and K^+ ions doping via a simple and fast one-step MS procedure. This tactic was highlighted by the fact that the phase and morphology controlling of g-C₃N₄ was achieved by simply adjusting the reaction temperature in the molten salt medium. Interestingly, the high-temperature liquid media of molten salt could not only help in the formation of tubular structure, but also

introduce some K^+ ions and cyano group defects in the framework of g-C₃N₄.

Therefore, the type-II homojunctions with embedded interfaces of triazine-based g-C₃N₄ nanoparticles and heptazine-based g-C₃N₄ microtubes were obtained. As anticipated, the prepared g-C₃N₄ homojunctions displayed a dramatically enhanced visible light photocatalytic H₂ evolution activity than single g-C₃N₄ phases and bulk g-C₃N₄. The immensely enhanced photocatalytic performance could be attributed to the synergistic effects of larger specific surface area, enhanced visible light absorption and efficient transfer and separation of photoinduced carriers. This work indicated that the construction of g-C₃N₄ phase homojunctions was an effective approach for boosting photocatalytic H₂ production, and thus opening a new avenue to design and construct highly efficient g-C₃N₄-based photocatalysts.

There are no conflicts to declare.

Acknowledgements

This research was supported by the Major Science and Technology Program for Water Pollution Control and Treatment “Research and Demonstration of Key Technologies for Water Ecological Environment Remediation and Water Safety Guarantee in Baiyangdian Lake and Daqing River Basin (Xiong'an New Area)” (2018ZX07110) - “Research and Engineering Demonstration of Groundwater Pollution Control and Remediation Technology for Typical Sewage Reclamation

Ponds in Tanghe Sewage Reservoir and Xiong'an New Area” (2018ZX07110005), and the National Natural Science Foundation of China (Grant Nos. 21571162).

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at

Journal Pre-proof

References

- [1] W.J. Ong, L.L. Tan, Y.H. Ng, S.T. Yong, S.P. Chai, Graphitic carbon nitride (g-C₃N₄)-based photocatalysts for artificial photosynthesis and environmental remediation: Are we a step closer to achieving sustainability? *Chem. Rev.* 116 (2016) 7159-7329.
- [2] M.S. Zhu, S. Kim, L. Mao, M. Fujitsuka, J.Y. Zhang, X.C. Wang, T. Majima, Metal-free photocatalyst for H₂ evolution in visible to near-infrared region: Black phosphorus/graphitic carbon nitride, *J. Am. Chem. Soc.* 139 (2017) 13234-13242.
- [3] L.H. Yao, D. Wei, Y.M. Ni, D.P. Yan, C.W. Hu, Surface localization of CdZnS quantum dots onto 2D g-C₃N₄ ultrathin microribbons: Highly efficient visible light-induced H₂-generation, *Nano Energy* 26 (2016) 248-256.
- [4] S.W. Cao, J.X. Low, J.G. Yu, M. Jaroniec, Polymeric photocatalysts based on graphitic carbon nitride, *Adv. Mater.* 27 (2015) 2150-2176.
- [5] Y. Li, X.Y. Wu, J. Li, K. Wang, G.K. Zhang, Z-scheme g-C₃N₄@CsxWO₃ heterostructure as smart window coating for UV isolating, Vis penetrating, NIR shielding and full spectrum photocatalytic decomposing VOCs, *Appl. Catal. B: Environ.* 229 (2018) 218-226.
- [6] H.J. Yu, R. Shi, Y.X. Zhao, T. Bian, Y.F. Zhao, C. Zhou, G.I.N. Waterhouse, L.Z. Wu, C.H. Tung, T.R. Zhang, Alkali-assisted synthesis of nitrogen deficient graphitic carbon nitride with tunable band structures for efficient visible-light-driven hydrogen evolution, *Adv. Mater.* 29 (2017) 1605148.

- [7] Z.X. Zeng, H.T. Yu, X. Quan, S. Chen, S.S. Zhang, Structuring phase junction between tri-s-triazine and triazine crystalline C_3N_4 for efficient photocatalytic hydrogen evolution, *Appl. Catal. B: Environ.* 227 (2018) 153-160.
- [8] R.C. Pawar, S. Kang, J.H. Park, J.H. Kim, S. Ahn, C.S. Lee, Room-temperature synthesis of nanoporous 1D microrods of graphitic carbon nitride (g- C_3N_4) with highly enhanced photocatalytic activity and stability, *Sci. Rep.* 6 (2016) 1-14.
- [9] K. Schwinghammer, M.B. Mesch, V. Duppel, C. Ziegler, J. Senker, B.V. Lotsch, Crystalline carbon nitride nanosheets for improved visible-light hydrogen evolution, *J. Am. Chem. Soc.* 136 (2014) 1730-1733.
- [10] Y.F. Li, R.X. Jin, Y. Xing, J.Q. Li, S.Y. Song, X.C. Liu, M. Li, R.C. Jin, Macroscopic foam-like holey ultrathin g- C_3N_4 nanosheets for drastic improvement of visible-light photocatalytic activity, *Adv. Energy Mater.* 6 (2016) 1601273.
- [11] W.B. Wang, Z. Shu, J. Zhou, T.T. Li, P. Duan, Z.L. Zhao, Y.G. Tan, C. Xie, S. Cui, Halloysite-derived mesoporous g- C_3N_4 nanotubes for improved visible-light photocatalytic hydrogen evolution, *Appl. Clay Sci.* 158 (2018) 143-149.
- [12] B. Liu, L.Q. Ye, R. Wang, J.F. Yang, Y.X. Zhang, R. Guan, L.H. Tian, X.B. Chen, Phosphorus-doped graphitic carbon nitride nanotubes with amino-rich surface for efficient CO_2 capture, enhanced photocatalytic activity, and product selectivity, *ACS Appl. Mater. Interfaces* 10 (2018) 4001-4009.

- [13] S.P. Wang, C.J. Li, T. Wang, P. Zhang, A. Li, J.L. Gong, Controllable synthesis of nanotube-type graphitic C₃N₄ and their visible-light photocatalytic and fluorescent properties, *J. Mater. Chem. A* 2 (2014) 2885-2890.
- [14] Z.W. Tong, D. Yang, Y.Y. Sun, Y.H. Nan, Z.Y. Jiang, Tubular g-C₃N₄ isotype heterojunction: Enhanced visible-light photocatalytic activity through cooperative manipulation of oriented electron and hole transfer, *Small* 10 (2016) 4093-4101.
- [15] G.G. Zhang, A. Savateev, Y.B. Zhao, L.N. Li, M. Antonietti, Advancing the $n \rightarrow \pi^*$ electron transition of carbon nitride nanotubes for H₂ photosynthesis, *J. Mater. Chem. A* 5 (2017) 12723-12728.
- [16] S.E. Guo, Z.P. Deng, M.X. Li, B.J. Jiang, C.G. Tian, Q.J. Pan, H.G. Fu, Phosphorus-doped carbon nitride tubes with a layered micro-nanostructure for enhanced visible-light photocatalytic hydrogen evolution, *Angew. Chem. Int. Ed.* 55 (2016) 1830-1834.
- [17] M. Wu, J. Zhang, B.B. He, H.W. Wang, R. Wang, Y.S. Gong, In-situ construction of coral-like porous P-doped g-C₃N₄ tubes with hybrid 1D/2D architecture and high efficient photocatalytic hydrogen evolution, *Appl. Catal. B: Environ.* 241 (2019) 159-166.
- [18] Z. Mo, H. Xu, Z.G. Chen, X.J. She, Y.H. Song, P.C. Yan, L. Xu, Y.C. Lei, S.Q. Yuan, H.M. Li, Self-assembled synthesis of defect-engineered graphitic carbon nitride nanotubes for efficient conversion of solar energy, *Appl. Catal. B: Environ.* 225 (2018) 154-161.

- [19] D.M. Chen, J.J. Yang, H. Ding, Synthesis of nanoporous carbon nitride using calcium carbonate as templates with enhanced visible-light photocatalytic activity, *Appl. Surf. Sci.* 391 (2017) 384-391.
- [20] Q. Han, B. Wang, Y. Zhao, C.G. Hu, L.T. Qu, A graphitic-C₃N₄ “seaweed” architecture for enhanced hydrogen evolution, *Angew. Chem. Int. Ed.* 54 (2015) 11433-11437.
- [21] J.H. Sun, J.S. Zhang, M.W. Zhang, M. Antonietti, X.Z. Fu, X.C. Wang, Bioinspired hollow semiconductor nanospheres as photosynthetic nanoparticles, *Nat. Commun.* 3 (2012) 1-7.
- [22] G.G. Zhang, L.H. Lin, G.S. Li, Y.F. Zhang, A. Savateev, S. Zafeiratos, X.C. Wang, M. Antonietti, Ionothermal synthesis of triazine-heptazine based Co-frameworks with apparent quantum yields of 60% at 420 nm for solar hydrogen production from "sea water", *Angew. Chem. Int. Ed.* 57 (2018) 9372-9376.
- [23] L.W. Hu, Y. Song, S.Q. Jiao, Y.J. Liu, J.B. Ge, H.D. Jiao, J. Zhu, J.X. Wang, H.M. Zhu, D.J. Fray, Direct conversion of greenhouse gas CO₂ into graphene via molten salts electrolysis, *ChemSusChem* 9 (2016) 588-594.
- [24] H.Q. He, J. Yin, Y.X. Li, Y. Zhang, H.S. Qiu, J.B. Xu, T. Xu, C.Y. Wang, Size controllable synthesis of single-crystal ferroelectric Bi₄Ti₃O₁₂ nanosheet dominated with {001} facets toward enhanced visible-light-driven photocatalytic activities, *Appl. Catal. B: Environ.* 156 (2014) 35-43.

- [25] J.N. Zhao, L. Ma, H.Y. Wang, Y.F. Zhao, J. Zhang, S.Z. Hu, Novel band gap-tunable K-Na co-doped graphitic carbon nitride prepared by molten salt method, *Appl. Surf. Sci.* 332 (2015) 625-630.
- [26] K. Schwinghammer, B. Tuffy, M.B. Mesch, E. Wirnhier, C. Martineau, F. Taulelle, W. Schnick, J. Senker, B.V. Lotsch, Triazine-based carbon nitrides for visible-light-driven hydrogen evolution, *Angew. Chem. Int. Ed.* 52 (2013) 2435-2439.
- [27] T. Liang, J.Y. Li, F. Liang, J.K. Wang, S.S. Li, H.J. Zhang, S.W. Zhang, Molten salt synthesis of tetragonal carbon nitride hollow tubes and their application for removal of pollutants from wastewater, *Appl. Catal. B: Environ.* 225 (2018) 307-313.
- [28] Y.P. Wang, Y.K. Li, W. Ju, J.C. Wang, H.C. Yao, L. Zhang, J.S. Wang, Z.J. Li, Molten salt synthesis of water-dispersible polymeric carbon nitride nanoseaweeds and their application as luminescent probes, *Carbon* 102 (2016) 477-486.
- [29] N. Li, J. Zhou, Z.Q. Sheng, W. Xiao, Molten salt-mediated formation of g-C₃N₄-MoS₂ for visible-light-driven photocatalytic hydrogen evolution, *Appl. Surf. Sci.* 430 (2018) 218-224.
- [30] Y.L. Li, S.X. Ouyang, H. Xu, X. Wang, Y.P. Bi, Y.F. Zhang, J.H. Ye, Constructing solid-gas-interfacial fenton reaction over alkalinized-C₃N₄ photocatalyst to achieve apparent quantum yield of 49% at 420 nm, *J. Am. Chem. Soc.* 138 (2016) 13289-13297.

- [31] V.W. Lau, I. Moudrakovski, T. Botari, S. Weinberger, M.B. Mesch, V. Duppel, J. Senker, V. Blum, B.V. Lotsch, Rational design of carbon nitride photocatalysts by identification of cyanamide defects as catalytically relevant sites, *Nat. Commun.* 7 (2016) 12165.
- [32] Z.F. Chen, S.C. Lu, Q.L. Wu, F. He, N.Q. Zhao, C.N. He, C.S. Shi, Salt-assisted synthesis of 3D open porous g-C₃N₄ decorated with cyano groups for photocatalytic hydrogen evolution, *Nanoscale* 10 (2018) 3008-3013.
- [33] G.G. Liu, G.X. Zhao, W. Zhou, Y.Y. Liu, H. Pang, H.B. Zhang, D. Hao, X.G. Meng, P. Li, T. Kako, J.H. Ye, In situ bond modulation of graphitic carbon nitride to construct p-n homojunctions for enhanced photocatalytic hydrogen production, *Adv. Funct. Mater.* 26 (2016) 6822-6829.
- [34] Z.A. Lan, G.G. Zhang, X.C. Wang, A facile synthesis of Br-modified g-C₃N₄ semiconductors for photoredox water splitting, *Appl. Catal. B: Environ.* 192 (2016) 116-125.
- [35] K. Wang, Q. Li, B.S. Liu, B. Cheng, W.K. Ho, J.G. Yu, Sulfur-doped g-C₃N₄ with enhanced photocatalytic CO₂-reduction performance, *Appl. Catal. B: Environ.* 176-177 (2015) 44-52.
- [36] S.Z. Hu, X. Chen, Q. Li, F.Y. Li, Z.P. Fan, H. Wang, Y.J. Wang, B.H. Zheng, G. Wu, Fe³⁺ doping promoted N₂ photofixation ability of honeycombed graphitic carbon nitride: The experimental and density functional theory simulation analysis. *Appl. Catal. B: Environ.* 201 (2017) 58-69.

- [37] J.C. Wang, C.X. Cui, Y. Li, L. Liu, Y.P. Zhang, W.N. Shi, Porous Mn doped g-C₃N₄ photocatalysts for enhanced synergetic degradation under visible-light illumination, *J. Hazard. Mater.* 339 (2017) 43-53.
- [38] L.S. Zhang, N. Ding, M. Hashimoto, K. Iwasaki, N. Chikamori, K. Nakata, Y.Z. Xu, J.J. Shi, H.J. Wu, Y.H. Luo, D.M. Li, A. Fujishima, Q.B. Meng, Sodium-doped carbon nitride nanotubes for efficient visible light-driven hydrogen production, *Nano Res.* 11 (2018) 2295-2309.
- [39] M.M. Zhang, C. Lai, B.S. Li, D.L. Huang, G.M. Zeng, P. Xu, L. Qin, S.Y. Liu, X.G. Liu, H. Yi, M.F. Li, C.C. Chu, Z. Chen, Rational design 2D/2D BiOBr/CDs/g-C₃N₄ Z-scheme heterojunction photocatalyst with carbon dots as solid-state electron mediators for enhanced visible and NIR photocatalytic activity: Kinetics, intermediates, and mechanism insight, *J. Catal.* 369 (2019) 469-481.
- [40] Z.Q. Liang, B.T. Sun, X.S. Xu, H.Z. Cui, J. Tian, Metallic 1T-phase MoS₂ quantum dots/g-C₃N₄ heterojunctions for enhanced photocatalytic hydrogen evolution, *Nanoscale*, 11 (2019) 12266-12274.
- [41] X.F. Yang, L. Tian, X.L. Zhao, H. Tang, Q.Q. Liu, G.S. Li, Interfacial optimization of g-C₃N₄-based Z-scheme heterojunction toward synergistic enhancement of solar-driven photocatalytic oxygen evolution, *Appl. Catal. B Environ.* 244 (2019) 240-249.

- [42] L. Pan, S.B. Wang, J.W. Xie, L. Wang, X.W. Zhang, J.J. Zou, Constructing TiO₂ p-n homojunction for photoelectrochemical and photocatalytic hydrogen generation, *Nano Energy* 28 (2016) 296-303.
- [43] F. Dong, Z.W. Zhao, T. Xiong, Z.L. Ni, W.D. Zhang, Y.J. Sun, W.K. Ho, In situ construction of g-C₃N₄/g-C₃N₄ metal-free heterojunction for enhanced visible-light photocatalysis, *ACS Appl. Mater. Interfaces* 5 (2013) 11392-11401.
- [44] M.C. Liu, D.W. Jing, Z.H. Zhou, L.J. Guo, Twin-induced one-dimensional homojunctions yield high quantum efficiency for solar hydrogen generation, *Nat. Commun.* 4 (2013) 2278.
- [45] S.Y. Tan, Z.P. Xing, J.Q. Zhang, Z.Z. Li, X.Y. Wu, J.Y. Cui, J.Y. Kuang, J.W. Yin, W. Zhou, Meso-g-C₃N₄/g-C₃N₄ nanosheets laminated homojunctions as efficient visible-light-driven photocatalysts, *Int. J. Hydrogen Energy* 42 (2017) 25969-25979.
- [46] B.Y. Ye, X.X. Han, M.D. Yan, H.H. Zhang, F.N. Xi, X.P. Dong, J.Y. Liu, Fabrication of metal-free two dimensional/two dimensional homojunction photocatalyst using various carbon nitride nanosheets as building blocks, *J. Colloid Interface Sci.* 507 (2017) 209-216.
- [47] H.H. Liu, D.L. Chen, Z.Q. Wang, H.J. Jing, R. Zhang, Microwave-assisted molten-salt rapid synthesis of isotype triazine-/heptazine based g-C₃N₄ heterojunctions with highly enhanced photocatalytic hydrogen evolution performance, *Appl. Catal. B: Environ.* 203 (2017) 300-313.

- [48] X.F. Liu, N. Fechler, M. Antonietti, Salt melt synthesis of ceramics, semiconductors and carbon nanostructures, *Chem. Soc. Rev.* 42 (2013) 8237-8265.
- [49] L.H. Lin, H.H. Ou, Y.F. Zhang, X.C. Wang, Tri-s-triazine-based crystalline graphitic carbon nitrides for highly efficient hydrogen evolution photocatalysis, *ACS Catal.* 6 (2016) 3921-3931.
- [50] F. Dong, M.Y. Ou, Y.K. Jiang, S. Guo, Z.B. Wu, Efficient and durable visible light photocatalytic performance of porous carbon nitride nanosheets for air purification, *Ind. Eng. Chem. Res.* 53 (2014) 2318-2330.
- [51] D. Dontsova, S. Pronkin, M. Wehle, Z.P. Chen, C. Fettkenhauer, G. Clavel, M. Antonietti, Triazoles: A new class of precursors for the synthesis of negatively charged carbon nitride derivatives, *Chem. Mater.* 27 (2015) 5170-5179.
- [52] W.X. Zheng, N.B. Wong, W.Z. Wang, G. Zhou, A.M. Tian, Theoretical study of 1, 3, 4, 6, 7, 9, 9b-heptaazaphenalene and its ten derivatives, *J. Phys. Chem. A* 108 (2004) 97-106.
- [53] R. Hailili, C.Y. Wang, E. Lichtfouse, Perovskite nanostructures assembled in molten salt based on halogen anions KX (X = F, Cl and Br): Regulated morphology and defect-mediated photocatalytic activity, *Appl. Catal. B: Environ.* 232 (2018) 531-543.
- [54] W. Iqbal, C.Y. Dong, M.Y. Xing, X.J. Tan, J.L. Zhang, Eco-friendly one-pot synthesis of well-adorned mesoporous g-C₃N₄ with efficiently enhanced visible light photocatalytic activity, *Catal. Sci. Technol.* 7 (2017) 1726-1734.

- [55] V.W. Lau, V.W. Yu, F. Ehrat, T. Botari, I. Moudrakovski, T. Simon, V. Duppel, E. Medina, J.K. Stolarczyk, J. Feldmann, V. Blum, B.V. Lotsch, Urea-modified carbon nitrides: enhancing photocatalytic hydrogen evolution by rational defect engineering, *Adv. Energy Mater.* 7 (2017) 1602251.
- [56] D.J. Martin, K.P. Qiu, S.A. Shevlin, A.D. Handoko, X.W. Chen, Z.X. Guo, J.W. Tang, Highly efficient photocatalytic H₂ evolution from water using visible light and structure-controlled graphitic carbon nitride, *Angew. Chem. Int. Ed.* 53 (2014) 9240-9245.
- [57] J.S. Zhang, M.W. Zhang, C. Yang, X.C. Wang, Nanospherical carbon nitride frameworks with sharp edges accelerating charge collection and separation at a soft photocatalytic interface, *Adv. Mater.* 26 (2014) 4121-4126.
- [58] Y.X. Li, H. Xu, S.X. Ouyang, D. Lu, X. Wang, D.F. Wang, J.H. Ye, In situ surface alkalized g-C₃N₄ toward enhancement of photocatalytic H₂ evolution under visible-light irradiation, *J. Mater. Chem. A* 4 (2016) 2943-2950.
- [59] X.G. Ma, Y.H. Lv, J. Xu, Y.F. Liu, R.Q. Zhang, Y.F. Zhu, A strategy of enhancing the photoactivity of g-C₃N₄ via doping of nonmetal elements: a first-principles study, *J. Phys. Chem. C* 116 (2012) 23485-23493.
- [60] J.G. Hou, H.J. Cheng, C. Yang, O. Takeda, H.M. Zhu, Hierarchical carbon quantum dots/hydrogenated- γ -TaON heterojunctions for broad spectrum photocatalytic performance, *Nano Energy* 18 (2015) 143-153.

- [61] S.W. Cao, B.J. Shen, T. Tong, J.W. Fu, J.G. Yu, 2D/2D heterojunction of ultrathin MXene/Bi₂WO₆ nanosheets for improved photocatalytic CO₂ reduction, *Adv. Funct. Mater.* 28 (2018) 1800136.
- [62] P.F. Xia, M. Antonietti, B.C. Zhu, T. Heil, J.G. Yu, S.W. Cao, Designing defective crystalline carbon nitride to enable selective CO₂ photoreduction in the gas phase, *Adv. Funct. Mater.* 29 (2019) 1900093.
- [63] C.B. Chen, C.X. Li, Y.J. Zhang, Y.J. Wang, J.Y. Lu, H.Q. Liu, W.W. Li, Cyano-rich mesoporous carbon nitride nanospheres for visible-light-driven photocatalytic degradation of pollutants, *Environ. Sci.: Nano.* 5 (2018) 2966-2977.
- [64] X.L. Liu, P. Wang, H.S. Zhai, Q.Q. Zhang, B.B. Huang, Z.Y. Wang, Y.Y. Liu, Y. Dai, X.Y. Qin, X.Y. Zhang, Synthesis of synergetic phosphorus and cyano groups (-C≡N) modified g-C₃N₄ for enhanced photocatalytic H₂ production and CO₂ reduction under visible light irradiation, *Appl. Catal. B: Environ.* 232 (2018) 521-530.

Figures

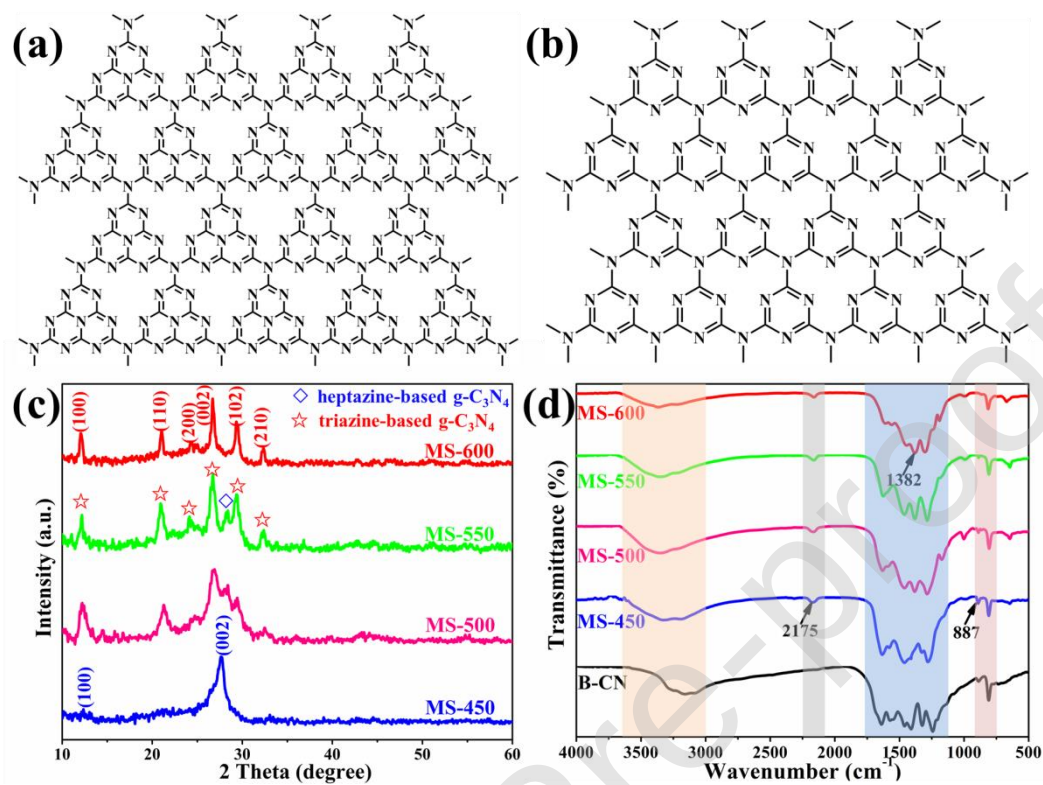


Fig. 1. (a) Tri-s-triazine (heptazine) and (b) Triazine structures of $g-C_3N_4$; (c) XRD patterns and (d)

FT-IR spectra of the as-synthesized samples.

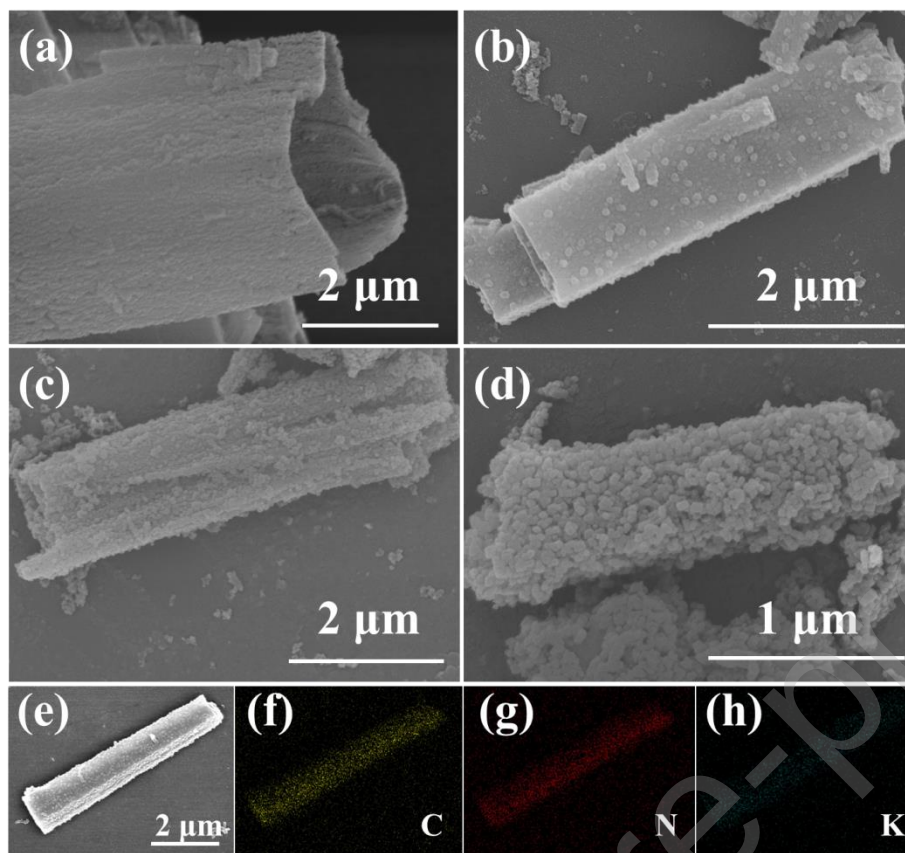


Fig. 2. Typical SEM images of (a) MS-450, (b) MS-500, (c) MS-550, and (d) MS-600; (e-h) the element mapping images of C, N and K for the MS-550.

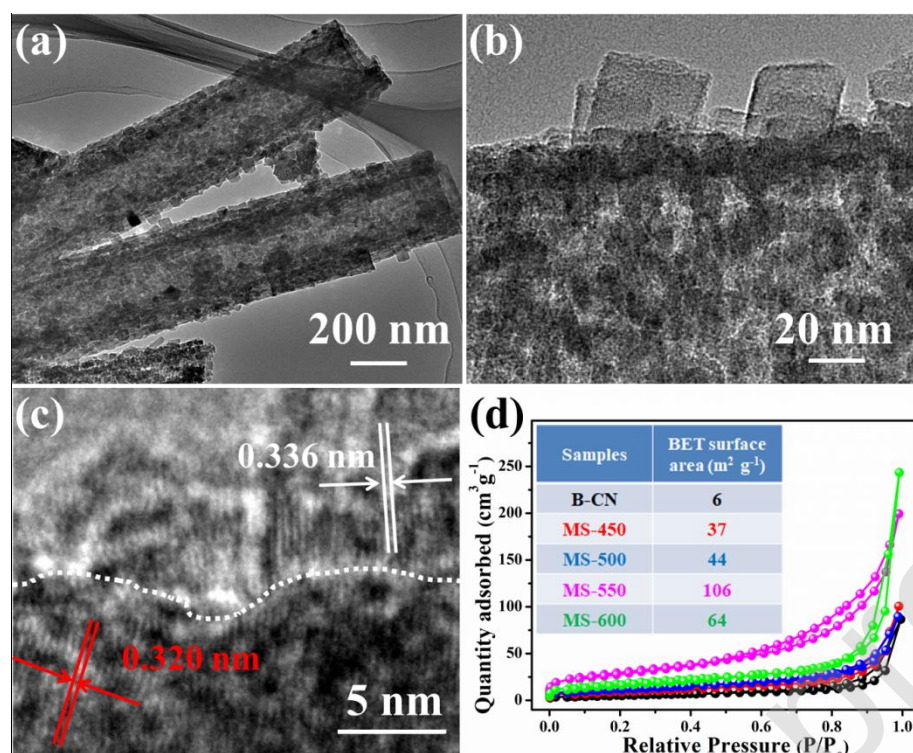


Fig. 3. (a, b) TEM and (c) HR-TEM images of MS-550; (d) N_2 adsorption-desorption isotherm curves and the BET surface areas of as-prepared samples.

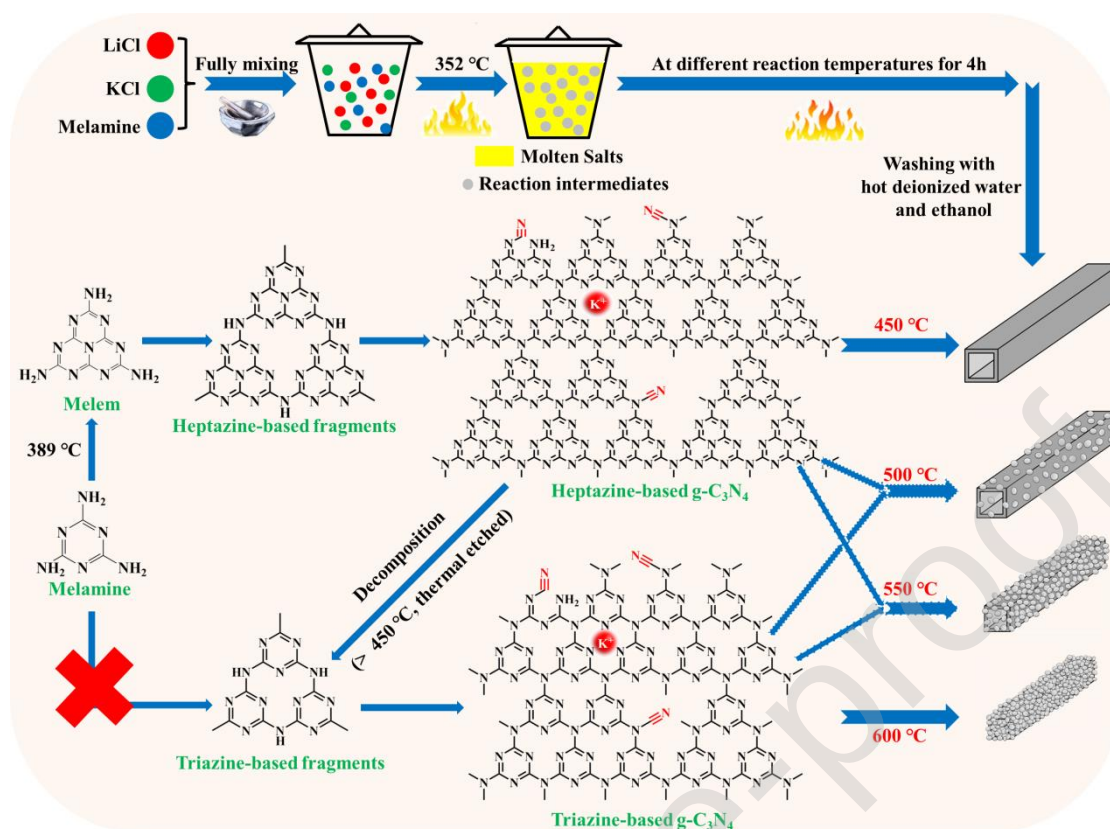


Fig. 4. Schematic illustration of the preparation process of MS-CN with different synthetic temperatures.

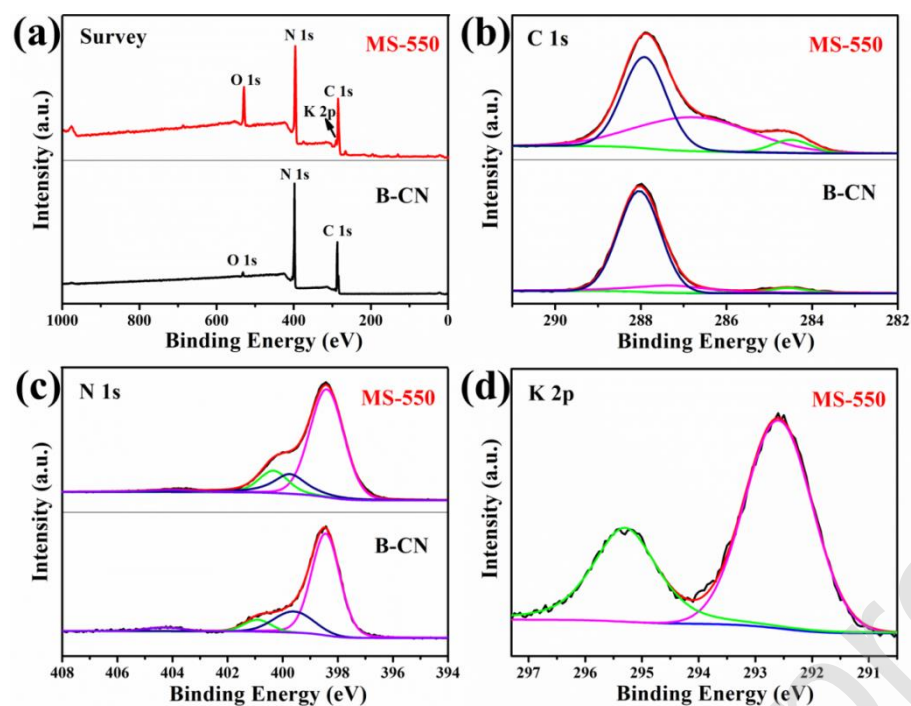


Fig. 5. XPS spectra: (a) survey, (b) C 1s, (c) N 1s of the B-CN and MS-550 samples, (d) K 2p of MS-550.

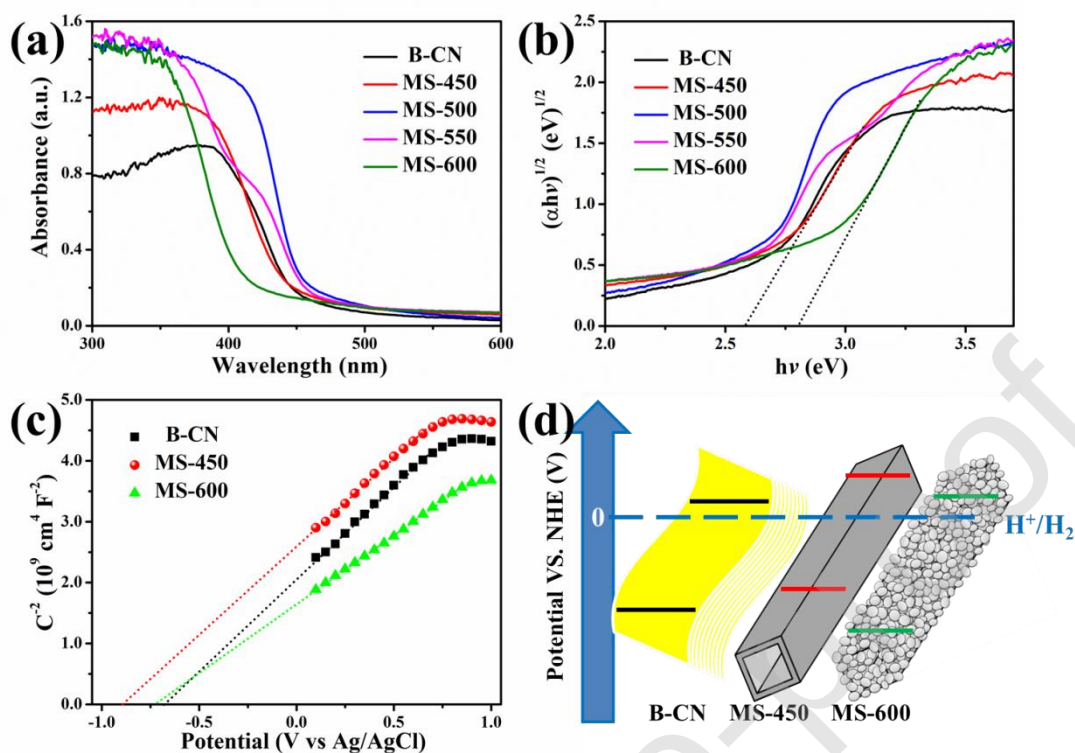


Fig. 6. (a) UV-vis diffuse reflectance spectra and (b) the curves between $(\alpha h\nu)^{1/2}$ and photon energy ($h\nu$) of the as-prepared samples; (c) Mott-Schottky plots and (d) schematic illustration of the band positions for B-CN, MS-450 and MS-600 samples, respectively.

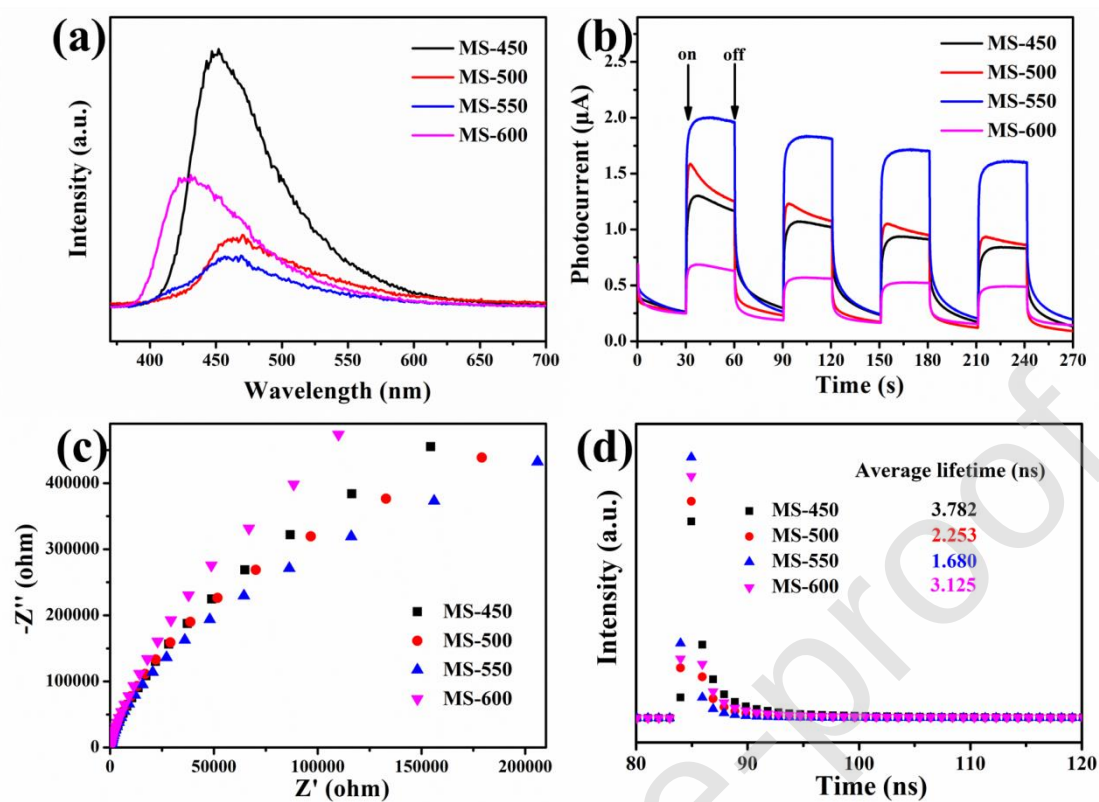


Fig. 7. (a) PL spectra, (b) Transient photocurrent responses, (c) EIS Nyquist plots and (d)

Time-resolved transient PL decay spectra of MS-450, MS-500, MS-550 and MS-600 samples.

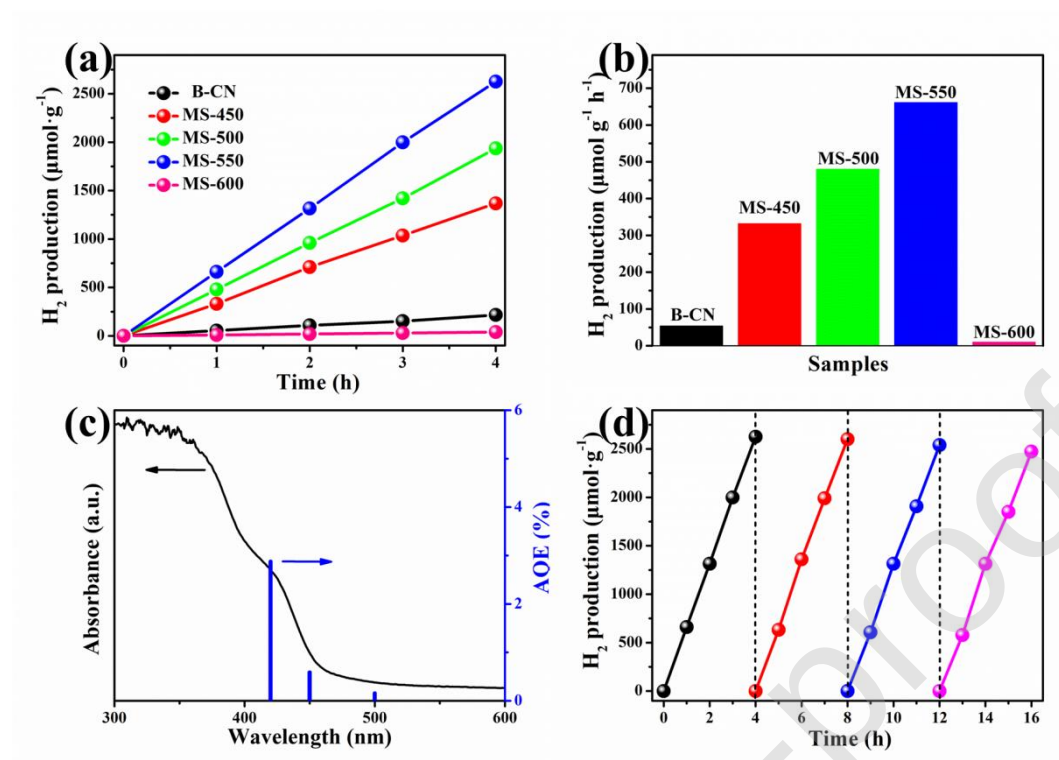


Fig. 8. (a) Time courses of visible light photocatalytic H_2 evolution; (b) A comparison of the photocatalytic H_2 production activities of B-CN, MS-450, MS-500, MS-550 and MS-600; (c) Wavelength-dependent AQE for H_2 evolution over MS-550 under different monochromatic light irradiation; (d) Cycling stability test for H_2 production of MS-550.

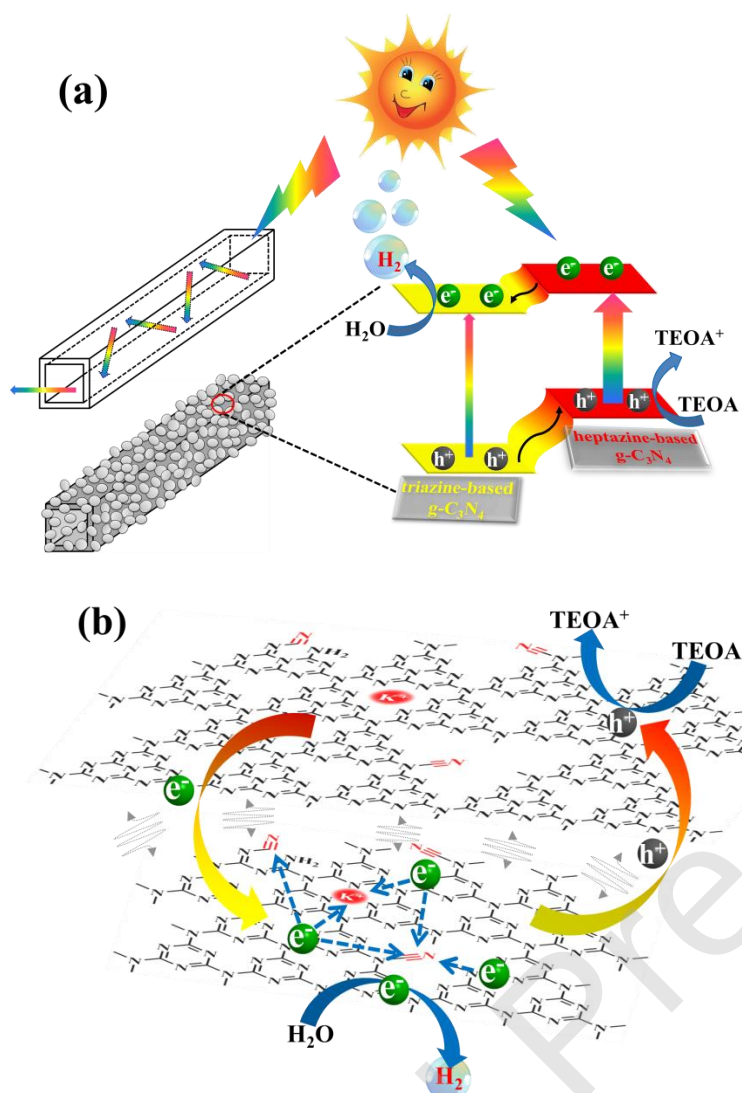


Fig. 9. Possible mechanism (a) and schematic expression (b) for the photogenerated charges transfer and induced photocatalytic H_2 evolution process of the triazine-/heptazine-based $g-C_3N_4$ homojunctions.